

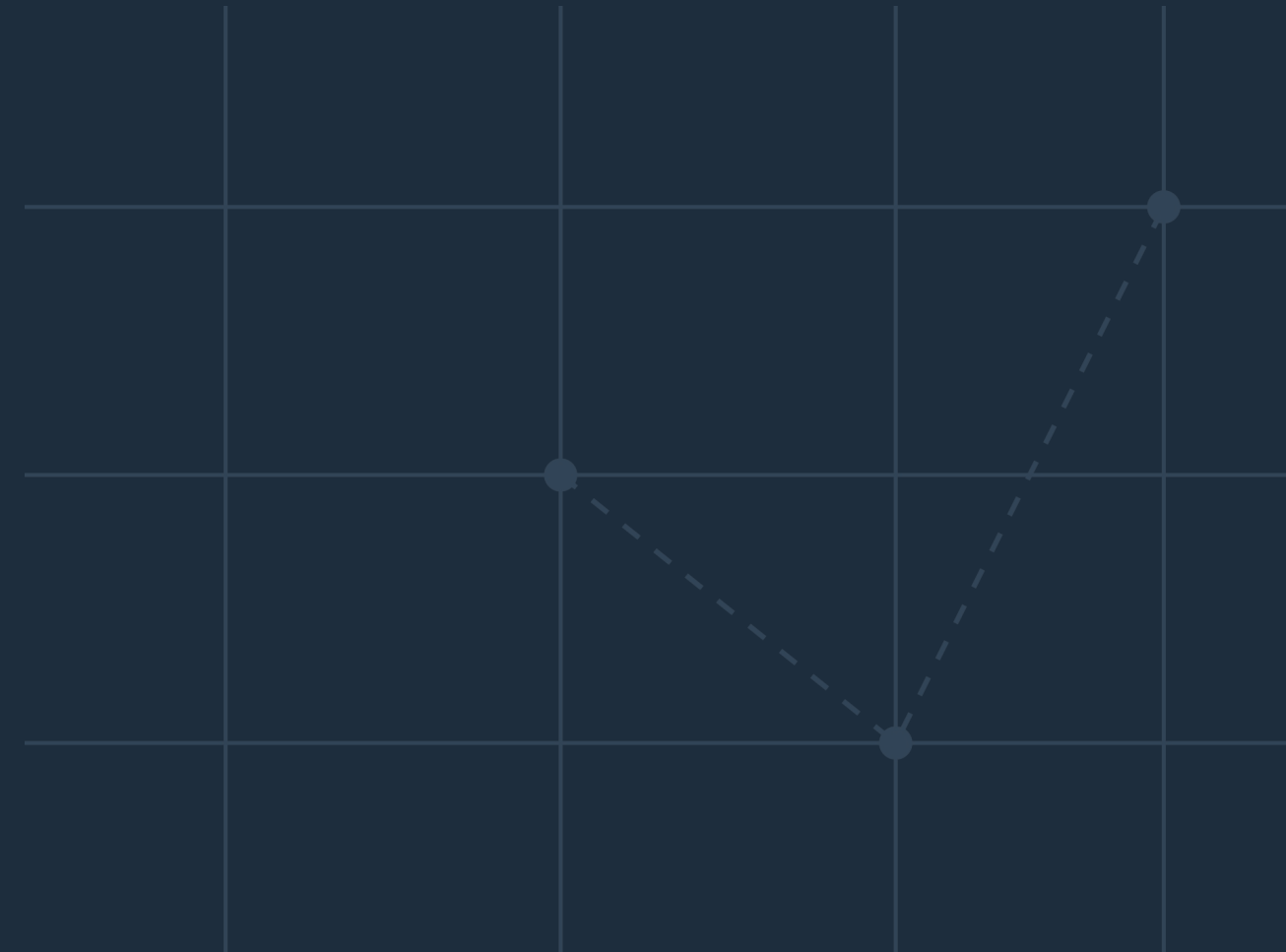
PROTECTIVE INTELLIGENCE
PLATFORM

Threat Detector

Protective Intelligence for Corporate Security

User — intelligence analysts protecting C-suite executives

Turns scattered open-source signals into ranked, sourced risk assessments



Why this Matters?



📄 Analysts juggle many disparate sources, mostly unstructured

👁️ Signal is sentiment and intent, not numbers and dates

🕒 Entire days lost to reading news; triage becomes guesswork

🔗 Agent-based because the task is iterative, tool-dependent, open-ended

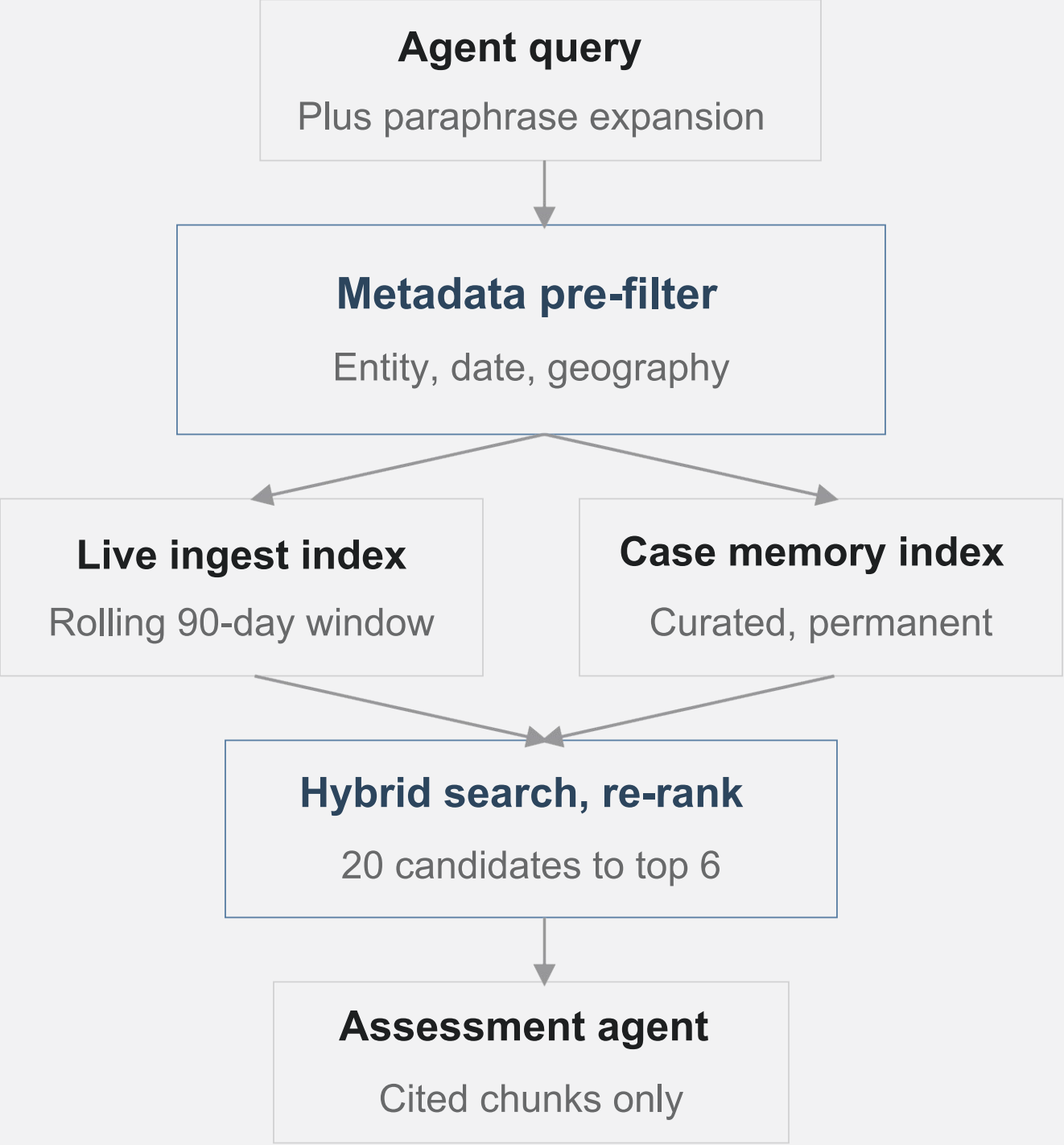
System Architecture

- 8 agents, hierarchical supervisor topology
- Orchestrator plans and delegates; holds no external tools
- 4 collection agents fan out in parallel, strictly read-only
- Assessment scores → recommendation proposes → reporting writes



Retrieval

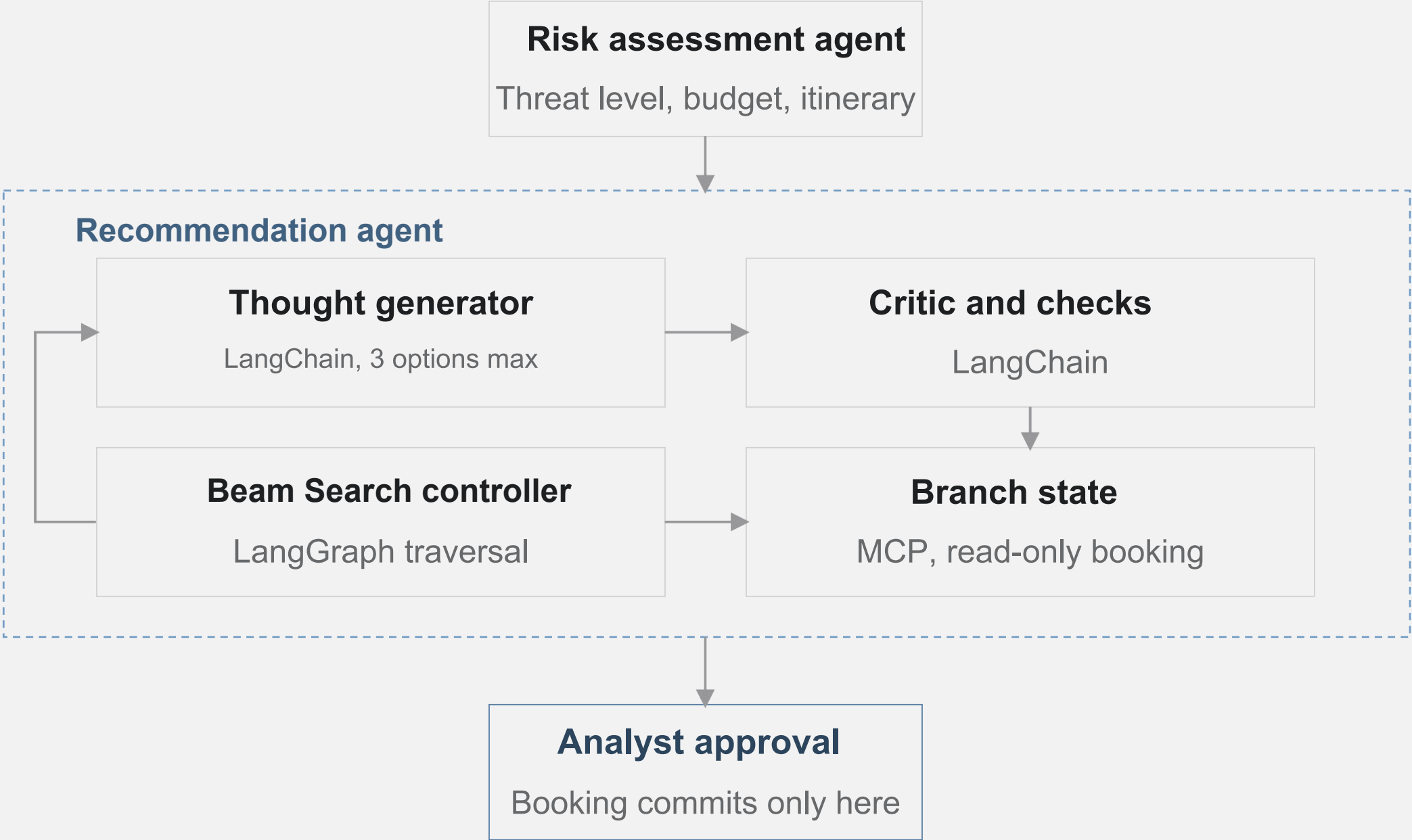
- 🗄️ Two indexes: rolling live ingest, permanent curated case memory
- 🎯 Metadata pre-filter runs first — entity, date, geography, access class
- 🔍 Hybrid dense + lexical search, re-rank 20 candidates down to 6
- 📄 Chunking varies by source type; social posts never split



Reasoning Structure

ToT Integration

- ToT nested inside the recommendation agent suggesting travel itineraries
- ✓ Controller (LangGraph) → Generator → critic (LangChain checks)
- 🗄 MCP carries branch state and wraps booking systems
- 🔒 Booking tools read-only during search — nothing commits until approval





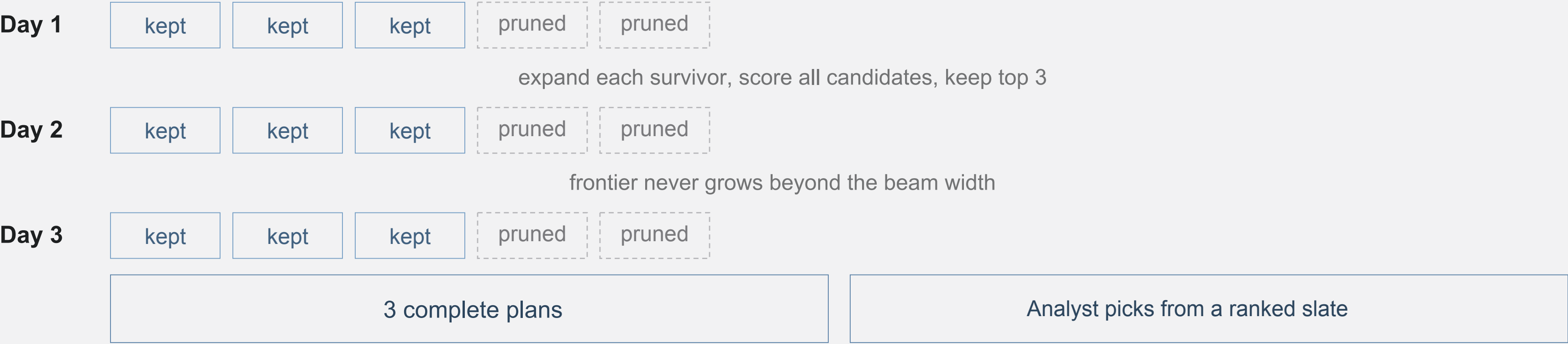
GUI Demo — recording to be added

Live walkthrough of threat triage and itinerary approval

Key Design Decisions



- DFS → beam search: avoids premature commitment on a shared budget
- CrewAI → LangChain (because CrewAI incompatible with Python 3.14)
- Open-source LLMs → frontier models (Sonnet orchestrator, Haiku specialists)
- Synthetic data for API cost, not privacy



Guardrails + Human Oversight



- 🔒 Structural: assessment agent can't retrieve; PII isolated in exposure agent
- ↪️ Output constraints: severity capped by corroboration
- ⚠️ "No evidence retrieved" ≠ "no threat identified"
- 🕒 Bounded loops with labeled-incomplete fallback
- 🛡️ Human gates: writes, spending, whitelist edits, itinerary commitment

Evaluation

- ✓ Groundedness — % of claims traceable to a retrieved document
- 🔍 Retrieval recall — seeded threat documents in a synthetic corpus
- 📄 Corroboration rate on high-severity findings
- ⚠️ Escalation rate — proxy for alert fatigue
- 🕒 Latency p95 and fallback success



Evaluation Metrics		
<i>A guardrail passes or fails on one item; a metric tells you whether the guardrails are calibrated. If escalation climbs gate.</i>		
metric	value	reading
groundedness rate	100%	target 95–100%; less = hallucination
escalation rate	80%	too high = alert fatigue; too low = gates catch nothing
corroboration rate	100%	0 of 1 high-severity claim(s) confidence-capped
stale / geo flags	0 / 0	guardrail catches this run
cap events	2	fallback success: True
volume	10 findings, 36 trace steps	

SOURCE & DOCUMENTATION

Github Repo

github.com/melvinchow/Threat_Detector_Agent

Full source, architecture docs, and eval harness

